

MaternaCare AI

Complete Project Execution Roadmap
FROM SETUP TO FINAL DEMO — STEP BY STEP

This roadmap breaks the entire project into 12 sequential phases and 50 concrete steps, from initial setup through final presentation. Follow the phases roughly in order; steps within Phases 4–8 can be split across team members and run in parallel where noted.

PHASE 0 | Setup

Week 1, Days 1-2

- 1. Set up a shared code repository** — create a GitHub repo, add all 3 members as collaborators, set up folders: /data, /notebooks, /models, /app, /docs.
- 2. Set up your Python environment** — Python 3.10+, virtual environment, install: scikit-learn, xgboost, pandas, numpy, matplotlib, deap (for GA), shap, imbalanced-learn.
- 3. Assign initial roles** per your team split (data/core model, explainability/clustering, interface/integration) — but review each other's code via pull requests so everyone understands the full system.
- 4. Create a shared doc** (Google Doc/Notion) to log decisions, issues, and results as you go — this becomes your final report's backbone.

PHASE 1 | Dataset Acquisition

Week 1, Days 2-4

- 5. Download the UCI Maternal Health Risk dataset** — confirm shape (1,014 rows, 6 features + label), check for missing values.
- 6. Download the public Anemia/CBC dataset** — confirm columns: Hemoglobin, MCH, MCHC, MCV, gender.
- 7. Do basic EDA on both datasets** — class distribution, feature ranges, correlations, missing values, outliers. Plot histograms per feature.
- 8. Document your EDA findings** in the shared doc — becomes your report's Data Understanding section.

PHASE 2 | Baseline Classifier

Week 1-2

9. **Split the maternal health data** into train/test (80/20), stratified by risk class.
10. **Train a baseline Random Forest** with simple settings — get accuracy, F1-score, confusion matrix.
11. **Train a baseline XGBoost** the same way.
12. **Benchmark against published results** (~88% accuracy reported in literature) — validate your pipeline before proceeding.
13. **Add trimester-wise segmentation** — infer or reasonably assign trimester tags (documented as a limitation if simulated), then evaluate the baseline separately per trimester.

PHASE 3 | Genetic Algorithm Optimization

Week 2-3

14. **Define your hyperparameter search space** for Random Forest/XGBoost (n_estimators, max_depth, min_samples_split, learning_rate).
15. **Set up DEAP** — define chromosome encoding, fitness function (cross-validated F1), selection/crossover/mutation operators.
16. **Run the GA optimization** — start small (population 20, 15 generations) to test end-to-end, then scale up (40-50 population, 30-40 generations).
17. **Compare GA-optimized model vs. baseline** on the same test set and metrics — document the improvement honestly.
18. **Repeat for both Random Forest and XGBoost** so you have two GA-optimized comparisons.

PHASE 4 | Longitudinal Trajectory Simulation

Week 3-4 (can overlap with Phase 3)

19. **Design your simulation logic** — generate 3-5 synthetic visits per patient with vitals drifting per pattern (Stable, Worsening, Sudden Deterioration, Improving), grounded in real clinical progression.
20. **Generate the simulated longitudinal dataset** — plot sample trajectories to confirm patterns look realistic, not random noise.
21. **Engineer trend features** — rate-of-change, rolling average, slope of vitals across visits.
22. **Train a trajectory-aware classifier** using trend features + latest vitals to predict current risk and trajectory pattern.

23. Document the simulation methodology clearly for transparency in your report.

PHASE 5 | Hybrid Clustering for Iron-Deficiency

Week 4-5 (parallel with Phase 4)

24. Preprocess the CBC/anemia dataset — scale/normalize features (clustering is sensitive to scale).

25. Run plain K-Means first as your baseline — try $K = 2, 3, 4$; use elbow method or silhouette score to pick K .

26. Implement GA-optimized centroid initialization — GA searches for good starting centroids, then K-Means refines (GKA-style hybrid).

27. Compare plain K-Means vs. GA+K-Means using silhouette score — this is your evidence for the hybrid module's value.

28. Interpret and label the clusters — check each cluster's average Hemoglobin to assign a clinical tier (Severe/Moderate/Low risk).

PHASE 6 | Explainability — SHAP

Week 5

29. Install and integrate SHAP with your GA-optimized classifier (TreeSHAP works directly with Random Forest/XGBoost).

30. Generate SHAP summary plots (overall feature importance) and individual prediction explanations (why this specific patient was flagged).

31. Sanity-check the explanations — do the top contributing features make clinical sense? Investigate if something looks off.

PHASE 7 | Recommendation Rules Engine

Week 5-6

32. Build your rules table (risk level × trajectory × trimester → action), grounded in WHO/NICE/ACOG thresholds.

33. Build the equivalent rules table for anemia clusters → supplement guidance.

34. Implement as a simple Python lookup — test against sample cases, including a 'default to caution' fallback for uncovered combinations.

PHASE 8 | Voice Interface

Week 6 (parallel with Phase 7)

35. Set up speech-to-text (Vosk for offline/lightweight, or Whisper for higher accuracy) — test with sample recorded vitals input.

36. Set up text-to-speech (gTTS or Coqui) — test reading back a sample recommendation.

37. Build a simple voice-input to voice-output loop as a standalone script before integrating into the full app.

PHASE 9 | Integration & Demo App

Week 6-7

38. Build a Streamlit (or simple React) app tying everything together: input vitals → GA-optimized prediction → SHAP explanation → rules-engine recommendation → voice output.

39. Add the anemia clustering module as a separate tab/section — CBC inputs → cluster/tier → supplement guidance.

40. Add the simulated escalation workflow — a mock alert (log entry/dummy notification) triggered on High Risk / Sudden Deterioration.

41. Test the full pipeline end-to-end with several sample cases, including edge cases (missing values, borderline readings).

PHASE 10 | Evaluation & Ablation Study

Week 7-8

- 42. Compile all comparison results** — baseline vs. GA-optimized (accuracy/F1/recall per class, especially High-risk recall), plain K-Means vs. GA+K-Means (silhouette score), with vs. without trend features.
- 43. Build clear comparison tables/charts** for your report and slides.
- 44. Write up your findings honestly** — a well-explained negative or mixed result is still good science.

PHASE 11 | Documentation & Paper Draft

Week 8-9

- 45. Update your project write-up doc** with actual results, replacing placeholder outcomes with real numbers.
- 46. Draft your paper/report** using your literature review, methodology, and evaluation sections as the backbone.
- 47. Prepare your final presentation slides** (similar structure to your Zeroth Review deck, now with real results).

PHASE 12 | Final Polish & Rehearsal

Week 9-10 (buffer)

- 48. Fix any bugs** found during full-system testing.
- 49. Rehearse your demo** — have a backup plan (screenshots/recorded video) in case live demo fails.
- 50. Do a mock review internally** — each member explains a different part to catch gaps before the real panel does.

Next step: begin Phase 0 + Phase 1 — repository setup and dataset acquisition — before moving into baseline modeling.